SHAHID: Telling Power Outages from Lost Meter Data at the Distribution Transformer

Anonymous submission
IEEE TSYP14 EnergyMind Tunisia challenge, Phase 1

***Abstract*—SHAHID is a probabilistic artificial intelligence trust layer between smart-meter data and dispatch decisions. When a group of meters under one distribution transformer stops reporting, it decides whether the customers lost power or whether only their data stopped. A meter on the low-voltage side of the transformer still sees the energy of customers whose data stopped, but not the energy of customers who lost power. SHAHID compares that unexplained energy with the consumption it expects from the silent customers, scores both causes with heavy-tailed likelihoods, and decides only when the evidence is decisive; otherwise it abstains and hands the case to a person with the evidence, a plain-language explanation and a tamper-evident log entry. All evidence is simulated on public German benchmark networks. With thresholds frozen on one network and each evaluation run once, the method called no labelled communication failure an outage on four other networks, missed at most one outage in a hundred, and decided about half to three quarters of the events. On data with no events, random single-meter data losses still produced about 15 to 33 false outage verdicts per transformer-year under the registered configuration; a post-hoc gate reduced this to 0.2 to 1.0. The evaluations also measure where the method fails: small silent groups, unmetered energy that follows the load, simultaneous events and hourly readings. The method requires an energy meter at each transformer and does not detect faults inside a single meter.**

***Index Terms*—abstention, advanced metering infrastructure, communication failure, distribution networks, energy balance, explainability, human oversight, outage detection, selective prediction, smart meters**

I. Introduction

The Tunisian utility, the Société Tunisienne de l'Électricité et du Gaz (STEG), is deploying smart meters. Press reports of a STEG briefing put the count at about 80,000 low-voltage (LV) meters in pilot zones by the end of March 2026, with a first phase of 450,000 electricity meters [1], [2]. STEG serves about 4.7 million subscribers, most of them residential LV customers [3]. In the summer of 2026 the same network went through rotating power cuts during a heatwave [4]–[6]. These facts come from secondary sources; they motivate the problem and enter no result.

As this advanced metering infrastructure (AMI) grows, operators will see groups of meters go silent. From the meter data alone they cannot tell an *outage*, which needs a crew, from a *communication failure*, which does not. A wrong call either wastes a truck roll or leaves customers without power and unattended. SHAHID addresses this one question for one LV neighbourhood under one medium-voltage/low-voltage (MV/LV) transformer, with 15-min readings. It is a probabilistic artificial intelligence (AI) trust layer between smart-meter data and dispatch decisions.

Hypothesis. A meter on the LV side of the transformer still measures the energy of customers whose data stopped, and no longer measures the energy of customers who lost power. If this difference is large compared with the uncertainty of the expected consumption, the two causes can be separated.

Contributions. All results are SIMULATED.

1) A transformer-side energy-balance test with abstention that, with thresholds frozen on one network, called no labelled communication failure an outage in every registered evaluation, on four networks never used for tuning. On data with no events, random single-meter data losses still produced about 15 to 33 false outage verdicts per transformer-year under the registered configuration; a post-hoc gate reduced this to 0.2 to 1.0 (Section X).
2) A measured operating envelope: the resolution limit, the reporting interval, and five conditions under which the method weakens or fails (Section X-C).
3) Governance measured, not assumed: an escalation table whose automatic rule was found to close real outages and was made provisional, operator load per transformer-day, a hash-chained decision log and an operator console preview (Sections VI–VII).
4) A failure mechanism measured and removed: a powered meter merged into an outage group can explain the whole residual; splitting silent groups by feeder removed the confident wrong verdicts it caused, 4 to 0 (Section X-D).
5) A verifiable evidence chain: protocols committed before each confirmatory run, negative results reported, and every number in this paper checked by code against a claims ledger (Section IX).

II. Related Work

Outages from AMI messages. Meters with outage notification send a "last gasp" message when power is lost, and utilities can "ping" meters to bound an outage [7]; pinging is also used to monitor grid-edge devices [8], and pings can enter a joint estimate of topology and outages [9]. SHAHID uses neither: its decision rests on interval energy alone, for the case where messages are absent or unreliable.

Outages from flows and forecasts. Sevlian et al. detect outages by combining line-flow measurements with load forecasts at the nodes of a distribution tree [10]; Wang et al. learn the normal distribution of smart-meter data per region [11]; a review surveys smart-meter analytics [12]. SHAHID shares

the principle of comparing a measured upstream flow with a forecast of downstream consumption, and applies it to a narrower question: which cause explains a group of meters that stopped reporting.

Deciding with a reject option. Chow characterised the trade-off between error and reject [13]; selective classification trades coverage for accuracy [14], [15], and reject strategies are compared in [16]. SHAHID's threshold rule is a constrained selection of this kind: it bounds missed and false outages, then maximises coverage. We make no novelty claim; these comparisons position the work.

III. Scope and Objectives

Scope. One LV neighbourhood under one MV/LV transformer, about 100 customers. Inputs: 15-min energy per customer meter with its arrival time, 15-min energy from one meter on the transformer's LV side, and static context (which meters share a line section or a data concentrator, installed photovoltaic (PV) power per customer). Output per silent group: OUTAGE, COMMUNICATION FAILURE or UNCERTAIN, with a probability, an explanation and a log entry.

Non-goals in Phase 1. No real data, no hardware, no deployed or real-time operation, no control action on the grid, no claim that the data represents Tunisia. Faults inside a single meter are tested but outside the validated scope (Section V-A). Current and power quality are out of scope: 15-min energy readings carry no waveform quantities.

Objectives, mapped to the brief.

- **G1, anomaly detection:** tell a genuine grid event from a data problem for silent groups; measure where this stops working (Sections V, X).
- **G2, confidence and explanation:** a probability, a plain-language explanation and an auditable record for every verdict (Section VI).
- **G3, escalation and governance:** rules for what the system may do alone, measured against their harm and their operator load (Section VII).
- **G4, shared learning across nodes:** what is shared, what stays local and why, with a measured proof of concept (Section VIII).

Appendix A maps every item of the brief to its status and evidence.

IV. Data and Anomaly Simulation

No STEG data is available to us, and no public dataset combines customer meter readings, a transformer meter, a communication layer and labelled causes. We therefore simulate the network and its measurement chain and label every event where it is injected. Table I states what is real and what is simulated, class by class.

The simulator writes four separate stores: hidden physical truth, the observed data a utility export could contain, static context, and labels. The detector reads only observed data and context; a test fails if it opens truth or labels. Outages and communication failures are drawn from the same sets of customers, so the pattern of silent meters carries no information about the cause. Every generator parameter is a configuration value in the repository, and the committed dataset regenerates with one command per folder.

V. Detection Method (G1)

Fig. 1 shows the pipeline. Meters whose readings for two consecutive intervals have not arrived form a *silent group* with a common onset t_0. At a decision time τ, using only data that had arrived by τ, the method computes over the window $w = [t_0, \tau)$

$$R = \sum_{t \in w} \left(T_t - \sum_{m \in K_t} E_{m,t} - g_0 - \hat{O}_t\right), \tag{1}$$

$$M = \sum_{t \in w} \sum_{m \in G} \left(\bar{E}_{m,t} + \rho_\ell \, \delta_m\right), \tag{2}$$

where T_t is the transformer's LV energy in interval t, K_t the meters whose reading for t has arrived, $E_{m,t}$ their energy, g_0 the median transformer-to-meters gap over the last day, $\hat{O}_t$ the expected energy of other missing meters, G the silent group, $\bar{E}_{m,t}$ the mean of the same 15-min slot over the previous 14 days, δ_m the deviation of meter m from that mean just before the onset, and ρ_ℓ a persistence factor fitted per lead ℓ. All energies are in kWh. If the customers still have power, $R \approx M$; if they lost it, $R \approx 0$.

Decision. With Student-t likelihoods of 4 degrees of freedom, scales s_O (outage) and s_C (communication failure) built from the fitted gap noise and forecast spread, and a prior of one half for each cause, the posterior probability of a communication failure is

$$p = \sigma\left(\ln f_\nu\left(\tfrac{R-M}{s_C}\right) - \ln s_C - \ln f_\nu\left(\tfrac{R}{s_O}\right) + \ln s_O\right), \tag{3}$$

where f_ν is the Student-t density and σ the logistic function. The verdict is COMMUNICATION FAILURE if $p \geq 1 - b$, OUTAGE if $p \leq b$, and UNCERTAIN otherwise, with abstention band $b = 0.03$. The decision is taken at the first of up to five decision times at which it is not UNCERTAIN.

Fitting and threshold selection. Noise and persistence parameters are fitted on each network's own fault-free history (training weeks); no labelled data is needed. The band b was chosen once, on development validation weeks, as the setting that abstains least while keeping missed outages at or below 0.05, false outages at or below 0.05 and fault-free false outages at or below 0.001 per meter-day (DESIGN CHOICE). Two earlier selection rules failed and are reported in the repository: one called 419 of 1051 fault-free silent groups outages, the other missed 0.366 of outages.

Why not a trained classifier. Under the same safety limits, logistic regression, gradient boosting and a small neural network trained on simulated labels missed 0.089, 0.179–0.196 and 0.232–0.250 of validation outages, against 0.036 for the fitted method; none had a feasible abstention band (Fig. 2). The fitted probabilistic parameters are therefore the trained models of this submission, by a measured choice.

TABLE I
WHAT IS REAL AND WHAT IS SIMULATED, CLASS BY CLASS

Element or anomaly class	Layer	Generation rule	Status (default value)
Networks, load and PV profiles	truth	Public German benchmark LV networks with their 2016 profiles (SimBench) [17]	REUSED, real benchmark data, not Tunisian
Electrical state	truth	AC power flow at every 15-min step (pandapower) [18]	REUSED solver
Customer diversity	truth	Per-customer daily shift, daily scale and correlated noise around the benchmark shapes, one parameter d	ASSUMPTION, results over its range
Physical outage	truth	A line is opened and the power flow is re-solved; isolated customers and their PV drop to zero	SIMULATED, from the event file
Real load change	truth	The load of a set of customers is scaled, then re-solved	SIMULATED, factor 0.3 to 2.5
Unmetered energy	truth	Added to the transformer series, constant or following the load shape	SIMULATED, none by default
Communication failure	observation	Readings of a concentrator, line section or random set stop; stored readings arrive after the link returns	SIMULATED, late delivery on
Random missing reading	observation	Each reading is lost independently	ASSUMPTION, probability 0.01
Delayed reading	observation	Each reading may arrive some intervals late	ASSUMPTION, probability 0.03
Meter and transformer-meter error	observation	Fixed gain, noise and quantisation per device	ASSUMPTION, gain sd 0.005 and 0.002
Single-meter faults (study only)	observation	Gain, bias and noise injected into one meter for seven days	SIMULATED, not in the dataset
Power-loss (last-gasp) message	observation	Optional message at outage onset	implemented, off in every result

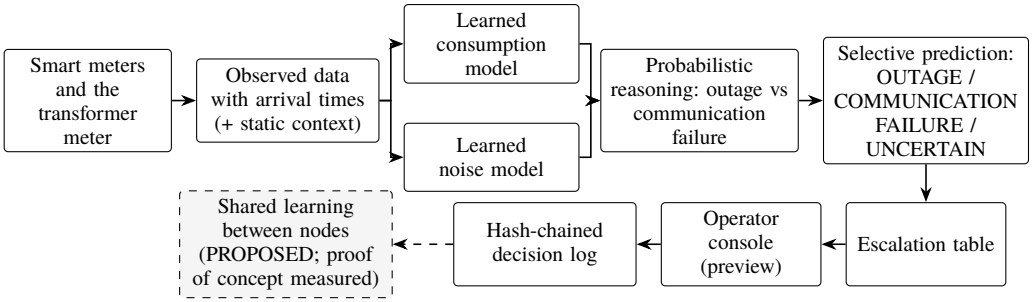


Fig. 1. System pipeline. Solid boxes are IMPLEMENTED and evaluated in simulation; dashed boxes are PROPOSED. The meter-count gate and the re-check of Section VII are evaluated in study code and are not yet executed by the escalation step.

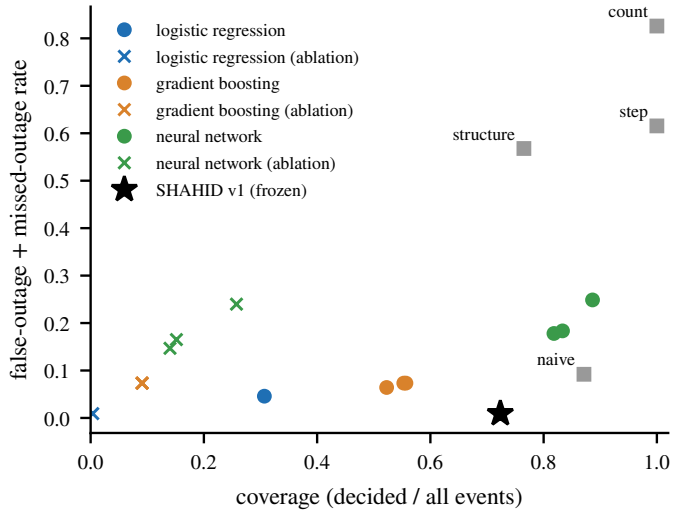


Fig. 2. Coverage against false-outage plus missed-outage rate on the held-out urban network, test weeks (SIMULATED, post-hoc). Dots: classifiers, one per seed; crosses: without energy-balance features; squares: simple baselines; star: the frozen method.

A. *Scope boundary: single-meter anomalies*

Silent groups were chosen first because the same symptom needs opposite responses and the transformer gives an independent physical witness. Faults inside one meter that keeps reporting (gain, bias, noise) are a different problem: the balance can show that some meter is wrong, not which one. We tested them. Injected gain errors, biases and noise on one meter were detected in at most 20% of cases; a per-meter regression of the transformer gap detected a gain of 0.5 in 0.347 of cases on a network never used before, and neighbour-voltage comparison detected at most 0.053. Single-meter fault detection is therefore outside the validated scope. The Phase 2 path (PROPOSED, not implemented): residual drift tracked per meter over weeks, comparison with electrically adjacent meters with field voltage data, and escalation to human review instead of automatic action.

VI. CONFIDENCE, EXPLANATION AND AUDIT (G2)

Confidence score. Every verdict carries the posterior p of (3). On the held-out network, probabilities below 0.1 (80 events) and above 0.9 (124 events) match the observed shares of communication failures (0.013 and 0.992). Over all 6210 held-out silent groups, mid-range values are not calibrated

TABLE II
ESCALATION TABLE (FIRST MATCHING RULE WINS)

Rule	Condition and action	Handled by
R1	Outage verdict on 20 meters or more: alert at once; operator decides on dispatch	human
R2	Outage verdict with probability 0.95 or more: alert with the evidence; operator confirms	human
R3	Communication verdict, probability 0.95 or more, at most 40 meters: provisional data-quality ticket; re-check the backlog of readings	automatic, provisional
R4	Any other communication verdict: ticket reviewed within the shift	human
R5	Abstention: review with the evidence	human

frequencies (expected calibration error 0.494), because most small groups are data losses decided under an equal prior. The console therefore labels mid-range values "current evidence strength, not validated accuracy".

Plain-language explanation. A fixed template is filled from the verdict record: how many meters went silent and when, the expected energy M, the unexplained energy R with the spread expected under each cause, the probability, the alternative cause and, when abstaining, the reason. A test fails if an explanation contains a number that is not in the record. Templates, not generated text, were chosen so that no explanation can state a number the system did not compute. The decision is interpretable by construction: it rests on two physical quantities in kWh, R and M, that an operator can check against the transformer reading; post-hoc explainers such as SHAP [19] were used only to inspect the comparison classifiers. Whether people can act on the explanations alone is registered as a study (protocol 07b) whose results are pending.

Auditable log. Each record (evidence, verdict, explanation, escalation rule) is appended to an append-only text log, one record per line; each entry stores the SHA-256 hash of the previous one, so any edit or deletion breaks the chain, and a test checks that a single edit is detected. Operator actions in the console go into a second chain of the same kind. The record answers what was detected, what was concluded and why.

VII. Escalation and Governance (G3)

Thresholds for automatic action. Table II is the escalation table, a configuration file (DESIGN CHOICE, not validated with operators). The only action taken without a person annotates data and opens a ticket; dispatching a crew or contacting customers always needs a person. As first written, rule R3 closed communication verdicts automatically. Measured on simulated events, it closed real outages without supervision in 0.9% to 1.8% of cases nominally and in 24.1% under 10 kW of load-shaped unmetered energy, so it was made provisional with a re-check.

Operator load. With 1% of readings lost at random and no events, version 1 sends 10.23 items per transformer-day to a person on the development network, almost all one- or two-meter silences. A meter-count gate (post-hoc variant) watches groups of fewer than 3 meters for one hour. On a network never used before, the gate that escalates instead of closing cut human items from 9.80 to 4.15 per transformer-day and false outage verdicts from 15.0 to 0.5 per transformer-year, with no outage closed silently among 350 whole-feeder outages. The gate that closes lost 16 of 350 outages on development data. **Implementation status:** the gate and the R3 re-check are evaluated in study code; neither is yet executed by the escalation step.

Operator interface. A validation and audit console (IMPLEMENTED as a preview, Streamlit) replays stored simulated days through the frozen pipeline: three queues that are never merged (outage awaiting approval, uncertain case, unconfirmed communication verdict), an evidence card per item with the alternative cause, a network map, and the hash-chained operator log with chain verification. Replay comes before live operation by design: verdicts and thresholds can be checked against known outcomes first. No operator has used it.

Over-trust and under-trust. Against over-trust: the alternative cause is shown with every verdict, a random sample of automatic closures goes to audit, and approvals given within seconds are flagged. Against under-trust: a track record per verdict class and overrides recorded with their outcome. These devices are implemented in the console; their effect on operators is not measured.

VIII. Shared Learning Across the Noosphere (G4)

What is shared, what stays local, and why. Shared between nodes: dimensionless fitted noise parameters, frozen decision thresholds, each site's measured resolution limit, and (PROPOSED) anonymous failure signatures such as "the gap follows the load" and operator-confirmed outcomes stripped of identifiers. Kept local: every meter reading, customer and meter identifiers and network topology, because interval consumption is personal data (DESIGN CHOICE).

What was tested. In a proof of concept, a new node received another node's noise-shape parameters and fitted only its scale from d days of its own data. The gain was negligible: with one day of local data, the nominal interval covered 0.978 of fault-free residuals when fitted locally and 0.96 with shared parameters, and shared parameters gave more fault-free outage verdicts (9 against 4). Local calibration is fast, so sharing noise parameters is the wrong mechanism.

What did transfer. Thresholds frozen on one network kept the false-outage rate at 0.0 on labelled communication failures on the networks never used for tuning (Section X). The design (PROPOSED) therefore rests on a bulletin of frozen thresholds and anonymous failure signatures: a new node preloads them, abstention rules are only ever tightened by shared knowledge, and no raw data leaves a site. A coordinator that sees several neighbourhoods go silent at the same moment could also treat them as one upstream event, such as a rotating cut.

IX. Evaluation Protocol

Networks and splits. Thresholds were chosen on one semi-urban development network (104 customers). Each network

is split by week into training, validation and test weeks. A held-out urban network (111 customers) was evaluated once after the method and thresholds were frozen and the protocol committed; an untouched rural network (118 customers) once later; and two networks never used before (rural, 99 customers; semi-urban, 41 customers) once in a confirmatory batch whose protocol was committed before the run. Events occur about once per day, an artificial rate chosen to obtain enough events, not a field frequency.

Metrics in the brief's terms.

- *Recall:* outage recall and communication recall, the share of each cause called correctly.
- *False-positive rate:* the false-outage rate (communication failures called outages), and false outage verdicts per meter-day on fault-free data.
- *Missed detections:* the missed-outage rate (outages called communication failures or never detected).
- *Coverage:* the share of events that receive a verdict. Accuracy when deciding is never reported without it.
- *Latency:* intervals from silence to verdict. *Explanation quality:* faithfulness by test; usefulness by a registered people study (pending).

A labelled event matches a silent group with the same onset and a meter-set overlap (Jaccard index) of at least 0.5; otherwise it is missed.

Integrity. Every number in this paper has a row in a claims ledger naming its result file, command, commit and configuration hash; a checker compares the paper with the ledger and the result files, and every table number must appear verbatim in a generated results document. The checker cannot catch a wrong two-decimal rate between 0 and 1, and we say so.

X. Results

All results are SIMULATED.

A. Held-out network and baselines

Table III compares methods on identical observed inputs: 109 outages and 155 communication failures on the held-out network, test weeks, evaluated once.

Version 1 is the safest method, not the one that decides most often: the naive baseline decides 0.871 of events but calls 0.065 of communication failures outages. On development validation data, only version 1 (coverage 0.663) and the z-test (0.602) can meet the three safety limits at any setting. The Isolation Forest detects that something happened but cannot attribute the cause (AUC 0.41, 95% interval 0.341–0.484, where AUC is the area under the receiver operating characteristic curve). **Leak check:** replacing the transformer series with a copy shifted by seven days drops outage recall from 0.641 to 0.103; the decisions depend on the transformer signal.

B. Every frozen evaluation

Table IV lists every evaluation of the frozen method. The development row uses unseen weeks of the network on which thresholds were chosen, so it is not held-out data.

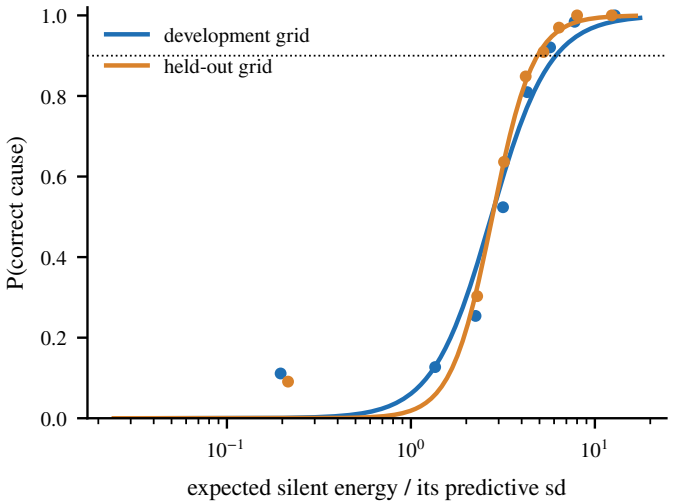


Fig. 3. Probability of a correct verdict against the ratio of expected silent energy to its uncertainty (SIMULATED; dots are binned shares, lines are logistic fits; dotted line at 0.9).

Accuracy when deciding was 1.0 on both never-used networks. The false-outage rates of Table IV (0.000) concern labelled communication failures. On fault-free data the two never-used networks gave 0.0004 and 0.0006 false outages per meter-day. On data with no events, random single-meter data losses still produced about 15 to 33 false outage verdicts per transformer-year under the registered configuration; a post-hoc gate reduced this to 0.2 to 1.0 (Section VII). Across the first three evaluations, outage recall was 0.42–0.67 and communication recall 0.54–0.75. The median decision latency is 2 intervals after the meters go silent.

C. Resolution limit and where the method weakens

The method is correct with probability 0.90 once the expected silent energy is about 6 times its predictive uncertainty: a signal-to-noise ratio (SNR) of 6.1 (5.2–7.3) on the development network and 4.9 (4.2–5.6) on the held-out network (Fig. 3). A single silent customer is below this limit, so it usually receives an abstention; the rare wrong verdicts on one- and two-meter groups are the fault-free false outages of Section VII.

Table V lists the conditions that degrade the method, on the development network with frozen thresholds, and the reporting-interval sensitivity on three networks with new seeds (post-hoc).

- **Main weakness.** The method treats the normal gap as constant; unmetered energy that follows the load and transformer-meter gain error both scale with load (Fig. 4).
- **Simultaneous events** merge into one silent group that looks like a communication failure.
- **Reporting interval.** Quarter-hour readings are safe; half-hour readings are inconclusive (missed outages 0.036, upper bound 0.052); hourly readings are outside the envelope, mostly because short events no longer suppress two readings.

TABLE III
HELD-OUT URBAN NETWORK, TEST WEEKS, EVALUATED ONCE

Method	Coverage	Accuracy when deciding	Outage recall	Comms recall	False outage	Missed outage	Fault-free false outages per meter-day
SHAHID version 1 (with abstention)	0.723	0.995	0.670	0.755	0.000	0.009	0.0010
Two-sided z-test[a]	0.636	0.994	0.615	0.645	0.000	0.009	0.0002
Likelihood ratio without abstention[a]	0.720	0.963	0.716	0.677	0.039	0.009	0.0038
Visible-share test[a]	0.720	0.963	0.725	0.671	0.045	0.000	0.0045
Naive share baseline[b]	0.871	0.943	0.890	0.774	0.065	0.028	0.0104
Step baseline (transformer drop)	1.000	0.682	0.752	0.632	0.368	0.248	0.0629
Count baseline (share of section silent)	1.000	0.545	0.826	0.348	0.652	0.174	0.0039
Structure baseline (silence pattern)[b]	0.765	0.604	0.624	0.348	0.394	0.174	0.0031
Isolation Forest [20] (added afterwards)	1.000	0.447	0.972	0.077	0.923	0.028	0.0057

[a]Coverage below 1 without an abstention band: these methods answer UNCERTAIN when the expected energy of the silent group is below a minimum chosen on validation weeks; the z-test also when both or neither cause fits. [b]The naive baseline has its own minimum-energy rule; the structure baseline abstains when the silent set is both a concentrator group and a line section. The count and step baselines and the Isolation Forest never abstain.

TABLE IV
EVERY EVALUATION OF FROZEN VERSION 1 (95% WILSON INTERVALS WHERE COMPUTED)

Network (run)	Coverage	False outage	Missed outage
Development, test weeks	0.697 (0.639–0.749)	0.000 (0.000–0.024)	0.000 (0.000–0.035)
Held-out urban (once)	0.723 (0.667–0.774)	0.000 (0.000–0.024)	0.009 (0.002–0.050)
Untouched rural (once)	0.494 (0.421–0.568)	0.000 (0.000–0.034)	0.000 (0.000–0.055)
Never used, rural, 99 meters (once)	0.670 (—)[a]	0.000 (—)[a]	0.009 (—)[a]
Never used, semi-urban, 41 meters (once)	0.527 (—)[a]	0.000 (—)[a]	0.000 (—)[a]

[a]Wilson interval not computed for the confirmatory runs.

TABLE V
CONDITIONS UNDER WHICH VERSION 1 WEAKENS OR FAILS

Condition	Coverage	False outage	Missed outage
Transformer-meter gain error, sd 2%	0.567	0.019	0.046
Load-shaped unmetered energy, 5 kW	0.534	0.034	0.122
Load-shaped unmetered energy, 10 kW	0.538	0.045	0.241
Outage and communication failure at once	0.761	0.016	0.373
10% of readings lost at random	0.490	0.000	0.135
Midday with doubled PV	0.528	0.000	0.013
Readings every 15 min (three networks)	0.629	0.002	0.006
Readings every 30 min	0.555	0.012	0.036
Readings every 60 min	0.421	0.021	0.245

- **Not sensitive to** constant unmetered energy, meter noise up to three times nominal, history length, PV share or concentrator layout. Coverage spans 0.484 to 0.663 over the range of the uncalibrated diversity parameter.

D. Further studies (protocols committed before running)

- **Merged powered meter.** A powered meter that goes silent at the same moment as an outage can join the outage group and supply its residual; on whole-feeder runs of the never-used rural network this happens 15.0 times per transformer-year. This is a different quantity from the 15.0 false outage verdicts of Section VII: both count 30 events over the same exposure, but in separate simulations (whole-feeder outages here, fault-free data there). Splitting silent groups by feeder removed every confident wrong verdict it caused (4 to 0) and stayed within the safety limits, at a price: abstention on random-meter communication failures rose from 0.405 to 0.973.
- **Negative results.** A prior that depends on group size missed 0.107 to 0.321 of outages, above the limit; evidence accounting version 2 missed more mid-interval outages than version 1 (0.257 against 0.117). Version 1 stays the registered method.

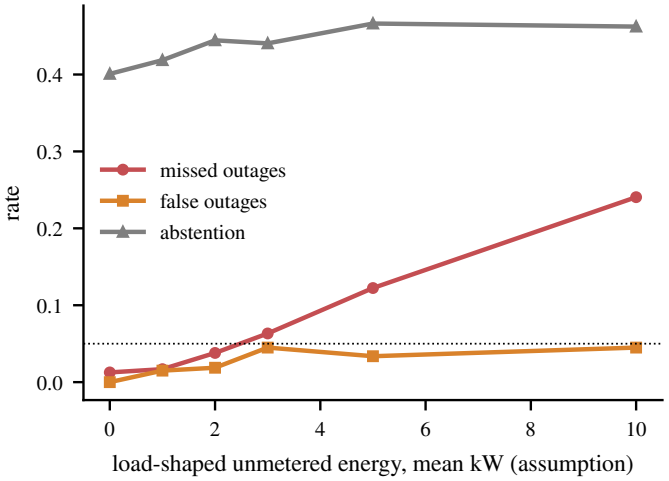


Fig. 4. Missed and false outages, and abstention, as load-shaped unmetered energy increases (SIMULATED, development network; dotted line at the 5% limit).

E. Deployment modes (Tunisia study)

A further study, on the same simulated networks with new seeds, varied the unknown Tunisian deployment conditions;

every value is an ASSUMPTION. Frozen version 1 transferred in 15 of 24 families and failed under 10% random loss and load-shaped unmetered energy. A transformer series delayed by a day costs only time. Without a transformer measurement, every energy-balance method answers UNSUPPORTED for 0.997 of episodes: the transformer meter is a prerequisite, not an option. With daily batch delivery the diagnosis becomes retrospective (median delay 10.25 h).

XI. Deployment and Business Case

All money is in Tunisian dinars (TND). No price was invented: a unit cost without a public source is shown as a formula (Table VI). The full case and its parameters are in the repository (`docs/BUSINESS_CASE.md`, `configs/business.yaml`).

Prerequisites. An LV energy meter at each transformer, read every 15 min, is required; it is a costed hardware line wherever absent. A head-end export of readings with arrival times, the line and zone register from the geographic information system (GIS), and the concentrator mapping complete the inputs.

Implementation. One pilot district the size of the STEG agency of Douz: about 24,400 customers and 149 transformer substations [21]. Phase A replays 90 days of the district's history with fits per transformer; Phase B runs in shadow. Go/no-go uses the frozen safety limits (missed and false outages at most 0.05, fault-free false outages at most 0.001 per meter-day); coverage and operator load are reported for STEG to judge.

Production and data protection. Decision and operator logs are append-only and hash-chained; crew dispatch always needs a person. Readings stay inside the utility. Organic Law 2004-63 applies, and a draft organic law of 132 articles, heard in parliament in February 2026, names data processing by artificial intelligence [22].

Continuous training. No labels are needed: each transformer's parameters are refitted on its own fault-free data every 90 days and after any topology change (DESIGN CHOICE); thresholds stay frozen. The fault-free false-outage rate per meter-day is the drift indicator.

Integration. Inputs from the AMI head-end, the transformer meter and the GIS; outputs to the outage-management or ticketing system and to the audit store.

Deployment cost. Compute and storage are negligible; operator review time dominates. Under version 1 a district would need 121.7 review hours per day at 5 min per item, at least 120486 TND per district-year; the gate that escalates cuts it to 51.5 hours, at least 51047 TND. These are floors: each hour is priced at the legal minimum wage of 2.713 TND [23], without employer charges or the qualification a grid operator needs. In staffing terms, 121.7 hours per day is about 15.2 eight-hour operator shifts, and 51.5 hours about 6.4. One operator around the clock reviews 288 items per day at 5 min each, that is at most 1.93 human items per transformer-day over 149 substations; the escalating gate gives 4.15, so the item rate must roughly halve. At 10% random data loss the load rises several-fold, so STEG's real data-loss level decides the budget.

Value. Crew time and service quality: every communication failure called correctly is a truck roll not sent, and either gate avoids 14.5 false outage verdicts per transformer-year on the rural network. Lost-sales arguments do not apply, because STEG sold electricity at 290.7 millimes per kWh in 2025 against an average production cost of 456.3 millimes [24]. SHAHID does not address theft or non-technical losses.

To verify with STEG. Which substations have an LV meter and how fast its data arrive; the customer reading interval and delivery cadence; store-and-forward behaviour; the real data-loss level; real event durations; labour and procurement prices; the acceptable operator load.

XII. Limitations

Each limitation is paired with its next action.

- **Simulation only.** The simulator and the detector were written by the same team, and energy conservation holds in the simulator by construction. *Next:* a retrospective test on one substation's historical data.
- **German benchmark networks.** No labelled Tunisian AMI data exists for this problem; the physics of the balance is the same, but profiles, topology and network sizes are not Tunisian. *Next:* recalibrate every fit and recheck every threshold on STEG data before any use.
- **Transformer meter required.** Without it the method cannot decide. *Next:* STEG's answer on transformer metering decides applicability.
- **Operator load.** Under the registered table it is the main cost. *Next:* wire the escalating gate into the escalation step and evaluate it with operators.
- **Events start on reading boundaries in simulation,** so missed-outage rates at 15 min are probably optimistic (HYPOTHESIS); late stored readings counted as reported energy push verdicts toward outage (a known flaw of version 1). *Next:* simulate sub-interval onsets and fix the accounting.
- **Event durations** of 1 to 4 hours are an assumption without a source; the re-check timeout depends on real durations. *Next:* field durations from STEG.
- **Not validated with people.** No operator has used the console and the explanation study is pending. *Next:* the registered people study and operator trials.

XIII. Conclusion and Phase 2 Plan

In simulation on five benchmark LV networks, a transformer-side energy balance with abstention separates power outages from communication failures for silent groups above its measured resolution limit, and it abstains below that limit instead of guessing. Every verdict carries its evidence, a probability, the alternative cause and a tamper-evident record. The evaluations also show where the method fails and what deployment would cost in operator time.

Phase 2 plan, in priority order.

TABLE VI
DEPLOYMENT COST FOR A DISTRICT OF 149 SUBSTATIONS (TND; FORMULAS WHERE NO PUBLIC UNIT COST EXISTS)

Line	Amount (TND)	Size from the simulation (ESTIMATE)
One-off		
Transformer LV meters	substations without a meter × (meter unit cost + installation)	share without a meter unknown
Integration	person-days × engineer day rate	head-end, ticketing and GIS adapters
Operator training	operators × cost per operator	—
Recurring per year		
Operator review time (main cost)	at least 120486.23 per district-year (version 1, 5 min per item)	121.7 hours per day at 5 min per item under version 1
Hosting	12 × monthly hosting cost	one process; 13.36 s of compute for one transformer-year
Storage	stored GB × price per GB-month	107.46 MB of readings per 1000 meter-years
Recalibration review	cycles per year × review days × day rate	every 90 days
Operator review, 1% data loss, 2 / 5 / 10 min per item: hours per district-day; at least TND per district-year		
Version 1	48.7 / 121.7 / 243.3 h; at least 48194.49 / 120486.23 / 240972.47	rural never-used network rates
Gate that escalates	20.6 / 51.5 / 103.1 h; at least 20418.66 / 51046.66 / 102093.31	no outage closed silently
Gate that closes	0.1 / 0.1 / 0.3 h; at least 53.75 / 134.38 / 268.76	lost 16 of 350 outages in development

1) Bring operator load to at most 1.93 human items per transformer-day, so that one operator around the clock covers a district of 149 substations (the escalating gate gives 4.15); wire the escalating gate and the communication recheck into the escalation step; operator trials of the console.
2) Run the registered people study on the explanations.
3) Fix the evidence accounting for late stored readings and sub-interval onsets; add a gap model that depends on load.
4) Single-meter faults: per-meter residual drift and neighbour comparison with field voltage data.
5) Shared learning through a bulletin of frozen thresholds and anonymous failure signatures.
6) Ask STEG the prerequisite questions of Section XI and run a retrospective test on one substation.

Repository. The submitted repository holds the code (MIT), the simulated dataset (ODbL 1.0, derived from SimBench, which is ODbL 1.0), the claims ledger, the protocols and the extended report (`docs/EXTENDED_REPORT.md`). Quickstart: `pip install -r requirements.lock`, `python -m pytest -q`, `python -m shahid.demo`. Figures and tables contain information from the SimBench database, made available under the ODbL 1.0.

REFERENCES

[1] African Manager, "Compteurs intelligents : 80 mille au compteur à fin mars 2026," Mar. 2026, accessed 13 October 2026. [Online]. Available: https://africanmanager.com/compteurs-intelligents-80-mille-au-compteur-a-fin-mars-2026/

[2] Webdo, "80.000 compteurs posés, 5 millions programmés : La STEG lance sa révolution intelligente," Mar. 2026, accessed 13 October 2026. [Online]. Available: https://www.webdo.tn/fr/?p=395508

[3] La Presse de Tunisie, "STEG et SONEDE : Les dettes clients dépassent 4,8 milliards de dinars," 18 February 2025, accessed 14 October 2026. [Online]. Available: https://www.lapresse.tn/2025/02/18/steg-et-sonede-les-dettes-clients-depassent-48-milliards-de-dinars/

[4] The New Arab, "As heatwave intensifies, Tunisia resorts to rolling power cuts to avoid total blackout," 17 July 2026, accessed 13 October 2026. [Online]. Available: https://www.newarab.com/news/amid-heatwave-tunisia-resorts-rolling-power-cuts

[5] African Manager, "Extreme heat, heavy losses and strange silence of STEG and Minister," 22 July 2026, accessed 13 October 2026. [Online]. Available: https://en.africanmanager.com/extreme-heat-heavy-losses-and-strange-silence-of-steg-and-minister/

[6] Webdo, "Délestages : La STEG annonce de nouvelles coupures ce mercredi dans plusieurs régions," 2 September 2026, accessed 13 October 2026. [Online]. Available: https://www.webdo.tn/fr/actualite/national/delestages-la-steg-annonce-de-nouvelles-coupures-ce-mercredi-dans-plusieurs-regions/403405/

[7] U.S. Department of Energy, Office of Electricity Delivery and Energy Reliability, "Advanced Metering Infrastructure and Customer Systems: Results from the Smart Grid Investment Grant Program," U.S. Department of Energy, Tech. Rep., 2016. [Online]. Available: https://www.energy.gov/sites/prod/files/2016/12/f34/AMI%20Summary%20Report_09-26-16.pdf

[8] C. Huang, C.-C. Sun, N. Duan, Y. Jiang, C. Applegate, P. D. Barnes, and E. Stewart, "Smart Meter Pinging and Reading Through AMI Two-Way Communication Networks to Monitor Grid Edge Devices and DERs," *IEEE Transactions on Smart Grid*, 2021, doi:10.1109/tsg.2021.3133952.

[9] A. Gandluru, S. Poudel, and A. Dubey, "Joint Estimation of Operational Topology and Outages for Unbalanced Power Distribution Systems," 2019, arXiv:1912.04543; related doi:10.1109/TPWRS.2019.2935401. [Online]. Available: https://arxiv.org/abs/1912.04543

[10] R. Sevlian, Y. Zhao, A. Goldsmith, R. Rajagopal, and H. V. Poor, "Distribution System Outage Detection using Consumer Load and Line Flow Measurements," 2016, arXiv:1503.08244, revised 17 Nov. 2016. [Online]. Available: https://arxiv.org/abs/1503.08244

[11] X. Wang, Z. Shi, B. Liu, W. Xiao, and S. Cheng, "Smart meter-based outage detection method for power distribution systems," *EAI Endorsed Transactions on Energy Web*, vol. 11, 2024, doi:10.4108/ew.5767.

[12] Y. Wang, Q. Chen, T. Hong, and C. Kang, "Review of Smart Meter Data Analytics: Applications, Methodologies, and Challenges," *IEEE Transactions on Smart Grid*, 2018, doi:10.1109/TSG.2018.2818167.

[13] C.-K. Chow, "On optimum recognition error and reject tradeoff," *IEEE Transactions on Information Theory*, vol. 16, pp. 41–46, 1970, doi:10.1109/TIT.1970.1054406.

[14] R. El-Yaniv and Y. Wiener, "On the Foundations of Noise-free Selective Classification," *Journal of Machine Learning Research*, vol. 11, no. 53, pp. 1605–1641, 2010.

[15] Y. Geifman and R. El-Yaniv, "Selective Classification for Deep Neural Networks," 2017, arXiv:1705.08500. [Online]. Available: https://arxiv.org/abs/1705.08500

[16] V. Franc, D. Prusa, and V. Voracek, "Optimal Strategies for Reject Option Classifiers," *Journal of Machine Learning Research*, vol. 24, no. 11, pp. 1–49, 2023.

[17] S. Meinecke, D. Sarajlić, S. R. Drauz, A. Klettke, L.-P. Lauven, C. Rehtanz, A. Moser, and M. Braun, "SimBench—A Benchmark Dataset of Electric Power Systems to Compare Innovative Solutions Based

on Power Flow Analysis," *Energies*, vol. 13, no. 12, p. 3290, 2020, doi:10.3390/en13123290.
[18] L. Thurner, A. Scheidler, F. Schäfer, J.-H. Menke, J. Dollichon, F. Meier, S. Meinecke, and M. Braun, "pandapower—An Open-Source Python Tool for Convenient Modeling, Analysis, and Optimization of Electric Power Systems," *IEEE Transactions on Power Systems*, vol. 33, no. 6, pp. 6510–6521, 2018, doi:10.1109/TPWRS.2018.2829021.
[19] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," in *Advances in Neural Information Processing Systems 30 (NIPS 2017)*, vol. 30, 2017. [Online]. Available: https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html
[20] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation Forest," in *2008 Eighth IEEE International Conference on Data Mining (ICDM)*, 2008, pp. 413–422, doi:10.1109/ICDM.2008.17.
[21] Managers, "À Douz, la Steg renforce un réseau électrique de 913 km," 15 July 2026, accessed 14 October 2026. [Online]. Available: https://managers.tn/2026/07/15/a-douz-la-steg-renforce-un-reseau-electrique-de-913-km/
[22] Business News, "ARP : audition autour du projet de loi sur la protection des données personnelles," 18 February 2026, accessed 14 October 2026. [Online]. Available: https://businessnews.com.tn/?p=1388983
[23] Managers, "Voici le nouveau SMIG en Tunisie," 30 April 2026, decree No. 67 of 2026. Accessed 14 October 2026. [Online]. Available: https://managers.tn/2026/04/30/voici-le-nouveau-smig-en-tunisie/
[24] ——, "Steg : l'électricité est vendue 290,7 millimes, mais coûte 456,3 millimes à produire," 29 June 2026, accessed 14 October 2026. [Online]. Available: https://managers.tn/2026/06/29/steg-lelectricite-est-vendue-2907-millimes-mais-coute-4563-millimes-a-produire/

Appendix A
Brief Coverage

Table VII maps every item of the brief to its status. The full table, with evidence paths, metrics and the Phase 2 plan for each gap, is `docs/BRIEF_COVERAGE.md`.

TABLE VII
BRIEF COVERAGE

Goal	Brief item	Status	Where
G1	Genuine grid event vs. data problem	IMPLEMENTED+MEASURED	Sections V, X
G1	Equipment faults, natural failures	IMPLEMENTED+MEASURED (loss of supply; cause not diagnosed)	Section X
G1	Degraded sensors, biased readings	TESTED — NOT DETECTED (negative result)	Section V-A
G1	Falsified readings	NOT ADDRESSED (Phase 2 design: persistent transformer-gap attribution)	Section V-A
G1	Corrupted readings	IMPLEMENTED+MEASURED (missing, delayed); wrong values TESTED — NOT DETECTED	Sections IV, V-A
G1	Voltage	TESTED — NOT USEFUL IN SIMULATION	Section V-A
G1	Current, power quality	OUT OF SCOPE (no waveform data)	Section III
G2	Confidence score	IMPLEMENTED+MEASURED (extremes reliable, mid range not)	Section VI
G2	Plain-language explanation	IMPLEMENTED+MEASURED (faithfulness); usefulness pending	Section VI
G2	Auditable log; what was detected, concluded and why	IMPLEMENTED+MEASURED	Section VI
G3	Thresholds for automatic action; human approval	IMPLEMENTED+MEASURED	Section VII
G3	Operator interface	IMPLEMENTED (preview; not used by operators)	Section VII
G3	Over-trust and under-trust	IMPLEMENTED (devices); effect not measured	Section VII
G4	What is shared, what stays local, and why	DESIGN-ONLY; proof of concept measured	Section VIII
Eval.	Precision/recall, false-positive rate, latency	IMPLEMENTED+MEASURED	Sections IX, X
Eval.	Explanation quality	IMPLEMENTED+MEASURED (faithfulness); people study pending	Section VI
Phase 1	Proposal, repository, dataset, anomaly documentation	IMPLEMENTED	Sections IV, XIII